E3a: the base's cofactor classes and the volcano study's IC yield effects

Result (measured, existing data, no new runs). Dividing each workload's IC relation yield by its exact class-cancelling ceiling `N0/r` removes the coordinate and horizontal heterogeneity and the pooled $\ell = 2$ floor/crater contrast. All four pre-registered gates in `PROTOCOL.md` (frozen in commit `23dbdb5c1`, before any N0 was regressed on a yield) pass by their frozen rules. Two secondary readings are weaker and are reported below, not folded into the verdict.

Scope: 24-bit p, `prime-abscissa:size=32`, MITM `negation_folded`, the six ordinary volcano classes of `research/isogeny-volcano-cost_20261017` (PROTOCOL v1.1): 800 workload curves, 12,800 verified runs. No security statement, and nothing here is about generators (E1) or a changed base (E3b).

Verdict wording (PROTOCOL §4): **yield follows the base's cofactor classes: slope 1.064 (cluster 95% CI [1.040, 1.090]), residual τ 0.3% horizontal and 0.2% coordinate (null 95th percentiles 1.4% and 2.5%).**

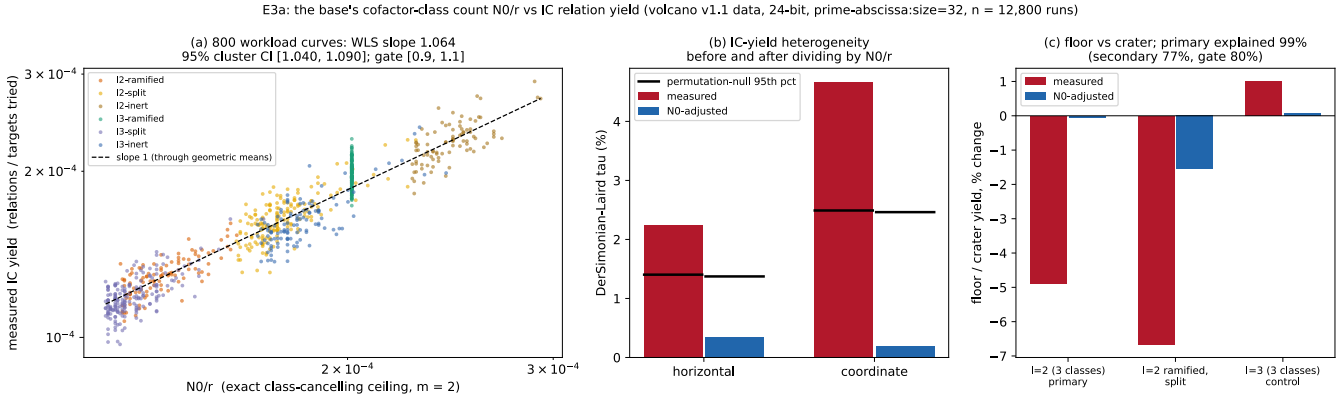


Figure 1: E3a

Gates

gate	frozen rule	measured	verdict
instrument	all 800 rebuilt bases hash to the recorded `points_sha256`; records match `MANIFEST.sha256`	800/800; manifest exact; 0 values capped at 1	pass
reproduction (not a gate)	`volcano-stats` reproduces the v1.1 IC rows	τ_h 2.25%, τ_model 4.67%, floor/crater 0.980 (volcano study: 2.25%, 4.67%, 0.980)	reproduced
G1 slope	WLS slope of ln yield on ln N0/r in [0.9, 1.1]	**1.064**, 95% [1.040, 1.090] (OLS 1.054)	pass
G2 horizontal τ	adjusted τ ≤ null p95	0.34% vs 1.37% (p = 0.57); was 2.25% vs 1.40%	pass
G3 coordinate τ	adjusted τ ≤ null p95	0.19% vs 2.46% (p = 0.96); was 4.67% vs 2.49%	pass
G4 $\ell = 2$ contrast	≥ 80% of ln(floor/crater) explained	measured 0.951 ⟶ adjusted 0.9995; **99%** explained, bootstrap 95% [71%, 158%]	pass

What the secondary readings say

- **The $\ell = 2$ contrast is −4.9%, not the −6.5% of the hypothesis.** The protocol's primary definition pools all three $\ell = 2$ classes with equal weight, and `l2-inert` (one crater curve) shows almost nothing. Its raw 95% interval is [−6.7%, −3.0%], so G4 was testable. For `l2-ramified` and `l2-split` alone (the two classes where the exploratory contrast was nominally significant) the raw contrast is −6.7% and N0 explains **77%**, under the 80% bar; the adjusted residual is −1.6% (90% interval [−3.7%, +0.6%], permutation p = 0.21). This is a secondary, so it does not change the verdict, but the "explains ≥ 80%" claim is solid for the pooled contrast and borderline for the two-class one. The intervals on the explained fraction are wide (the ratio of two small numbers).
- **Slope inside curves is at the edge of the band.** The class fixed-effect slope is 1.085 [1.003, 1.165] and the curve fixed-effect slope (models of one curve only, the coordinate effect) is 1.103 [1.008, 1.191]. The frozen gate is the pooled point estimate, which passes. The within-curve estimate is imprecise (its interval spans the band's edge); it is consistent with slope 1 and does not show a coordinate effect beyond N0.
- **$\ell = 3$ control.** Raw contrast +1.0% [−0.3%, +2.3%], adjusted +0.08%: nothing to explain (G4′s "not testable" rule applies to it).
- **Yield by cofactor h** (context): h = 1: 1.97e−4 (128 curves), h = 4: 1.62e−4 (368), h = 5: 1.18e−4 (188), h = 7: 1.61e−4 (116). `N0/r` ranges 1.24e−4 to 2.93e−4 (sd of ln = 0.21).

What it does not show

- `N0/r` is an upper bound, and a slope near 1 across six classes means the shortfall (duplicate sums, the negation fold) is roughly constant. The slight excess over 1 (1.06) is inside the pre-registered tolerance but is not explained here.
- The adjustment is by `N0/r` as computed for the *measured* base; it says yield follows it, not that a different base (`subgroup=filter|project`, E3b) would raise yield by the predicted factor. That is E3b.
- Per-class τ and the residual after adjustment are small, not zero; at this N a residual below about 1% is not distinguishable from none.
- The generator question (E0, E1) is not touched.
- Not a speed measurement, so no `benchmarks/` entry (nothing timed; counted yields only).

Method and files

- `src/bin/generator_n0.rs` (native Rust): `census` rebuilds each base with `prime_factor_base(&inst, 32)`, checks its hash, computes `exact_decomposition_ceiling(…, m = 2)` (and m = 3, unused); `regress` does G1 (WLS, OLS and fixed-effect variants, cluster bootstrap of the 200 curves, 2,000 replicates) and G4 (class-stratified contrast, curve bootstrap within class × level, 10,000-label permutation).
- `volcano-stats --n0 n0.json` (G2, G3): divides each run's relations by its workload's `N0/r` (numerator, so `(num, den)` stay exchangeable), then the unchanged heterogeneity tests with 2,000-permutation nulls. Seed 20261017 throughout; only `ic-mitm` and `rho-neg` records of the `spec` sessions (with `l3-split/spec-r2`) were passed, so the null percentiles differ from the volcano study's in the third digit.
- Outputs: `n0.json`, `n0.csv` (per workload), `workloads.csv` (yield and `N0/r` per workload), `regression.json`, `stats-unadjusted.{json,log}`, `stats-adjusted.{json,log}`; `plot.py` draws the figure from them.
- Reproduce:

```
cargo build --release --bin generator-n0 --bin volcano-stats
D=research/generator_effect_ic_20261019
./target/release/generator-n0 census --out $D/n0.json --csv-out $D/n0.csv
./target/release/generator-n0 regress --n0 $D/n0.json --out $D/regression.json --csv-out $D/workloads.csv
```

then `volcano-stats` as in the `argv` fields of `stats-*.json`. - Sources: sessions `research/isogeny-volcano-cost_20261017/sessions/*/spec` (`l3-split/spec-r2`); the volcano branch is `aburan28/youthful-ride-6u51pp-volcano-v2`, so this branch is based on it. - Graphs checked and left unchanged: the volcano study's figures (this adds none to them); no canonical scoreboard graph is affected.

Consequence for the plan

E3a's pre-registered success means the PLAN's working hypothesis stands for the three volcano effects: they are N0 effects. E3b (new factor-base arms at equal B) is the next test of whether the same lever raises yield.